Source and Relational Equation Recovery with WPM

Third sources, interactions, anchors, delays and missing histories

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Abstract

Equation recovery is an inverse problem whose answer depends on the observation map, admissible source class and available measurements. This study develops additive and nonlinear third-source inversion, signed-probe recovery with raw anchors, readjustment after a known source is disclosed, time-varying and delayed interaction fitting, latent-coordinate inversion and backward history reconstruction. Rank conditions and exact counterexamples separate identifiable parameters from observational equivalence classes. Worked trails show both successful inversion under stated assumptions and irreducible ambiguities. The objective is an auditable mathematical procedure rather than unrestricted symbolic discovery.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 State which recovery problem is being solved

The prerequisites are least squares, matrix rank, differential equations and Bayes' rule. Consider observations

$$Y_j = H(A_j, B_j, C_j; \gamma) + \epsilon_j. \tag{1.1}$$

Known sources A and B, hidden source C, observation parameters γ and noise all have distinct roles. "Known A" may mean a supplied equation with uncertain parameters, not a noiseless measured state. State exactly what is given.

Recovery can mean one of several non-equivalent tasks: reconstructing sampled C values; fitting coefficients in a specified C dictionary; choosing an operator or delay; identifying a function up to an equivalence class; forecasting the combined Y; or selecting the actual missing history. Accurate Y does not certify accurate C. Correct operator labels do not imply exact coefficients. Every returned equation should state its coordinate, units, domain, admissible grammar and unresolved alternatives.

Let $\mathcal{M}$ be a declared set of candidate models. An observation operator $\mathcal{O}$ maps a candidate latent object x to noiseless observations. Two candidates are observationally equivalent if $\mathcal{O}(x_1) = \mathcal{O}(x_2)$. Recovery from those observations can at most identify the equivalence class unless additional assumptions exclude members.

2 A non-identifiability result that applies to pockets too

Theorem 2.1 (Derived summaries cannot distinguish equal data)**.** *If $\mathcal{O}(x_1) = \mathcal{O}(x_2)$, every deterministic pocket or landscape extractor T gives the same result on these observations. In a statistical model, if $S = T(Y)$, then conditioning on (Y, S) gives the same information as conditioning on Y alone.*

Proof. The first claim is equality under function composition. For the second, S is measurable with respect to $\sigma(Y)$, so $\sigma(Y, S) = \sigma(Y)$. Conditional distributions and expectations given these sigma-fields agree, up to the usual null sets. □

Pockets can organize features, guide numerical search or preserve a useful statistic. This theorem does not say that every practical raw-data estimator already uses the data efficiently. It says that derived summaries are not extra independent observations. Multiplying a raw-data likelihood by a pocket likelihood as if independent generally counts the same evidence twice.

At a differentiable observation mean f_θ, the stacked Jacobian of raw observations and their deterministic summary is $[J; DT(f_\theta)J]$, which has the same column rank as J. Local information directions are not created by summary duplication. Nondifferentiable basin extraction does not invalidate the equal-data argument.

3 Linear source recovery and the role of anchors

Suppose $Y - b = A\theta + \epsilon$ with known baseline b and dictionary A. Signed probes give $s = Q(Y - b)$ and projected design QA.

Theorem 3.1 (Probe recovery with anchors)**.** *In a noiseless linear model, coefficients are identifiable from probes and anchor rows R exactly when*

$$\operatorname{rank} \begin{bmatrix} QA \\ RA \end{bmatrix} = p, \qquad p = \dim \theta. \tag{3.1}$$

Equivalently, $\ker(QA) \cap \ker(RA) = \{0\}$.

Proof. Two coefficient vectors give the same probes and anchors precisely when their difference lies in both nullspaces. The stacked linear map is injective exactly when its nullspace is trivial. □

Worked example 3.2 (Second differences lose a line)**.** At $u = (0, 1, 2, 3)$ let $C(u) = a + bu + cu^2$ and

$$Q = \begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{pmatrix}. \tag{3.2}$$

The probes give $(2c, 2c)$, so the projected design has rank one. Take measured anchors $C(0)$ and $C(1)$. If the packet is $(1, 1)$ and anchors are 2 and $11/2$, then $c = 1/2$, $a = 2$, and $b = 11/2 - 2 - 1/2 = 3$. Anchors restore both null directions.

If probes and anchors reuse noisy observations, fit them with their joint covariance or fit the raw observations. They cannot be treated as independent replicated measurements. For singular transformed covariance, retain exact zero-noise constraints and whiten only positive-variance directions.

For positive definite Σ, full-rank GLS has solution $(A^\top \Sigma^{-1} A)^{-1} A^\top \Sigma^{-1} (Y - b)$. With nuisance baseline columns D, identification of θ requires full column rank of the residualized whitened design $(I - \widetilde{D}\widetilde{D}^{+})\widetilde{A}$. A strong prior can stabilize a singular fit but cannot establish data-driven uniqueness.

4 Readjusting a fitted source when a known term arrives

Suppose a single-source line was fitted before learning that $Y = A_\theta + B$. For a fixed design X, unweighted least squares originally gives $\widehat{\theta}_{\text{old}} = (X^\top X)^{-1} X^\top Y$. After B is supplied,

$$\widehat{\theta}_{\text{new}} = \widehat{\theta}_{\text{old}} - (X^\top X)^{-1} X^\top B. \tag{4.1}$$

The correction is a projection of B into the fitted dictionary, not subtraction of B's symbolic coefficients from an unrelated equation. In a nonlinear or newly selected dictionary, refit the residual problem; the displayed identity need not hold.

Worked example 4.1 (An omitted quadratic changes both line coefficients)**.** Let $Y = 2 + 3u + u^2/2$ at $u = 0, 1, 2, 3$. Fitting only $(1, u)$ gives $\widehat{Y} = 3/2 + (9/2)u$. Once $B = u^2/2$ is known, refitting $Y - B$ gives $A = 2 + 3u$. The original intercept was wrong as well as the slope. The new A equation plus supplied B reconstructs Y exactly on these points.

If B is estimated from the same observations, residual uncertainty includes $\operatorname{Var}(Y - \widehat{B}) = \operatorname{Var}(Y) + \operatorname{Var}(\widehat{B}) - 2\operatorname{Cov}(Y, \widehat{B})$. A plug-in subtraction that ignores this covariance can exaggerate precision.

5 Third-source recovery: additive and calibrated multiplicative cases

For a genuinely additive calibrated map $Y = A + B + C + \epsilon$, one obtains $C = Y - A - B - \epsilon$. Subtracting known sources reconstructs a noisy C series, after which a declared C model can be fitted. If unmodeled interactions remain, the residual is not C alone.

For a calibrated map

$$Y = A + B + \lambda(A, B)C + \epsilon, \tag{5.1}$$

pointwise inversion is $\widehat{C} = (Y - A - B)/\lambda$ wherever $\lambda \neq 0$. With exact A,B,λ and additive noise variance σ^2, inverse noise variance is σ^2/λ^2. Near zero gain, a formally unique inverse is unstable. At zero gain the observation has no pointwise information about C.

Worked example 5.1 (A nonadditive source trail)**.** Suppose $\lambda = 1 + 0.4A$, $A = 1$, $B = 2$, and $Y = 3.7$. Then $\lambda = 1.4$ and $C = (3.7 - 1 - 2)/1.4 = 1/2$. If noise SD is 0.07, C's measurement SD from this inversion is 0.05. Changing the assumed gain changes both the recovered source and its uncertainty.

With several channels $Y_l = b_l + \lambda_l C + \epsilon_l$ and known covariance Σ, use $\widehat{C} = (\lambda^\top \Sigma^{-1} \lambda)^{-1} \lambda^\top \Sigma^{-1} (Y - b)$. Its variance is $(\lambda^\top \Sigma^{-1} \lambda)^{-1}$ under the stated model. Additional channels help only to the extent they add correctly modeled information.

6 Quadratic, multichannel and branch-valued inverses

For known α, β,

$$r = Y - A - B = \alpha C + \beta C^2. \tag{6.1}$$

When $\beta \neq 0$ and the discriminant is nonnegative, candidates are $C_\pm = (-\alpha \pm \sqrt{\alpha^2 + 4\beta r})/(2\beta)$. Both must be retained unless a domain constraint or additional measurement distinguishes them. Their observation derivatives are $\alpha + 2\beta C$; near zero derivative, inverse sensitivity becomes large.

Worked example 6.1 (An exact two-root ambiguity)**.** For $r = C + C^2$ and $r = 2$, the roots are 1 and -2. Both fit exactly. Choosing the positive root is justified only by an independently stated positivity constraint, not by better residual fit. A second oriented channel $z = C$ would distinguish them; an energy channel $z = C^2 + C$ would not.

For a vector map $H : \mathbb{R}^p \to \mathbb{R}^d$, full column rank of J_H supplies local identifiability under regularity through a locally invertible submap. It does not prove global uniqueness. For $p = 1$, nonzero derivative in at least one channel suffices locally, but looping or symmetric maps can still identify distinct distant sources.

In noisy data, a negative discriminant need not mean the physical model is impossible; it may mean noisy observations lie outside the exact noiseless image. Fit the observation likelihood with domain constraints rather than forcing complex-valued latent states. If exact constraints are asserted and violated beyond admitted error, report model inconsistency.

7 Unknown mixing laws and source gauge freedoms

Theorem 7.1 (Unrestricted hidden-source reparameterization)**.** *If H is unrestricted and g is an invertible transformation of C, the pair*

$$C' = g(C), \qquad H'(A, B, c') = H(A, B, g^{-1}(c')) \tag{7.1}$$

produces identical Y. Thus C's unrestricted equation is not identifiable from Y alone.

Proof. Substitute $C' = g(C)$ into H' and cancel $g^{-1}g$. Every observation, and hence every derived landscape, is unchanged. □

Even linear mixing has a scale ambiguity: $Y = A + B + \lambda C$ is unchanged by $C' = aC$, $\lambda' = \lambda/a$ for $a \neq 0$. Two unrestricted additive hidden sources have a larger freedom: $C' = C + h$ and $D' = D - h$ preserve $C + D$.

To remove such freedoms, specify calibrated gains, units, measured anchors, physical domains, independence or sparsity assumptions with identifiable consequences, or source-specific sensors. A regularized decomposition is useful only with its constraints stated. It is not a proof that the selected decomposition is the actual generator.

8 Recovering time-varying and delayed relations

Let a residual satisfy

$$r(t) = \sum_{j=1}^{J} \theta_j(t)\phi_j(A(t), B(t-\tau_j)), \qquad \theta_j(t) = \sum_{q=1}^{Q} \beta_{jq} b_q(t). \tag{8.1}$$

For each declared delay pattern, construct columns $b_q(t)\phi_j(A(t), B(t-\tau_j))$ and fit the coefficients. Sampling, delay interpolation, basis dimension and smoothness penalties are modeling choices. A single scalar snapshot can supply at most one scalar equation, so it cannot identify an arbitrary set of coefficients or slopes.

Worked example 8.1 (A coefficient and its slope)**.** Take $r(t) = (0.2 + 0.05t)(1 + \sin t)$. Where $1 + \sin t \neq 0$, divide to obtain $r(t)/(1 + \sin t) = 0.2 + 0.05t$. Two distinct noiseless times identify intercept and slope. One time yields a line of coefficient pairs. Near a zero denominator, fitting the original product form is numerically preferable to division.

Delay labels can be equivalent along a particular trajectory. If $B(t) = e^{ct}$, then $B(t-\tau) = e^{-c\tau}B(t)$; a freely estimated gain absorbs the delay. More time points on this same exponential do not identify gain and delay separately. Different excitation, independently fixed gain, or another channel can break the alias.

For differential relations, use an integral model when appropriate:

$$X(t_b) - X(t_a) = \int_{t_a}^{t_b} f(X(s), u(s); \theta)\, ds + \epsilon_{ab}. \tag{8.2}$$

Overlapping intervals create correlated errors. Integral fitting reduces some differentiation noise but introduces quadrature and noisy-regressor issues; it is not assumption-free.

9 Measurement precision, excitation and candidate selection

With independent replicate errors, averaging M readings divides noise covariance by M. With a common error component the covariance is $\Sigma_{\text{common}} + \Sigma_{\text{independent}}/M$. Repetition cannot break exact aliases in the mean design. To improve structural distinguishability, choose permitted conditions whose sensitivity matrices add independent directions.

In nonlinear recovery, separate three failure questions: was an adequate structure in the candidate grammar; did fitting reach a suitable solution; did the selection rule choose among observationally distinguishable candidates? Generating more structures cannot solve an information-theoretic alias. Finer numerical curvature cannot create a missing measurement direction.

For a fully specified probabilistic model, model probabilities satisfy

$$p(M \mid Y) \propto p(M) \int p(Y \mid \theta, M) p(\theta \mid M)\, d\theta. \tag{9.1}$$

This requires a likelihood, prior, parameter domain and model bank. Softmax-normalizing arbitrary residual or complexity scores only defines relative weights. It does not make them calibrated probabilities that an equation is true. Equivalent algebraic forms should not be counted as independent models without an explicit prior over that multiplicity.

10 Backward recovery of histories and causal memory

Let $X_{-H:-1}$ be a missing past and $Y_{0:T}$ observed later data. Under a declared state-space model with parameters θ, retrodiction uses

$$p(X_{-H:T}, \theta \mid Y_{0:T}) \propto p(\theta)p(X_{-H} \mid \theta) \prod_{j=-H}^{T-1} p(X_{j+1} \mid X_j, \theta) \prod_{j=0}^{T} p(Y_j \mid X_j, \theta). \tag{10.1}$$

Past anchors or memory channels enter as additional observations. Future relative to the missing interval is legitimate evidence for this retrospective task; it would be unavailable for a prospective forecast issued before it was observed.

Theorem 10.1 (Merged histories cannot be separated by identical later evidence)**.** *Suppose histories H_1, H_2 give the same conditional law for every observed future datum. If both have positive prior probabilities, posterior odds equal prior odds.*

Proof. Bayes' rule writes posterior odds as prior odds times the likelihood ratio. The latter is one by assumption. □

For deterministic noninjective dynamics, distinct earlier states can merge into one complete state and then have identical futures. A longer observation of the same merged state does not undo the loss. A delayed measured response may help if it stores information absent from that state description.

Worked example 10.2 (A signed memory channel identifies a branch probabilistically)**.** Suppose an observed squared state gives $x^2 = r^2$ and candidate pasts $x = \pm r$. An independent measured trace satisfies $z = bx + \epsilon$, $\epsilon \sim N(0, \sigma^2)$ with known b. With equal priors,

$$\log \frac{p(x = r \mid z)}{p(x = -r \mid z)} = \frac{2brz}{\sigma^2}. \tag{10.2}$$

To derive it, subtract the two Gaussian negative squared residuals. Taking $r = 0.6$, $b = 0.4$, $z = 0.22$, $\sigma = 0.1$ gives log odds 10.56. If b has an unidentified symmetric sign, integrating over that symmetry can restore equal branch probabilities. If the trace is $z = bx^2 + \epsilon$, both signs have identical likelihoods.

11 A practical branch-preserving smoother

Exact inverse roots are suitable for a noiseless known transition map. With uncertain transitions or noisy boundary data, use distributions over states and parameters. Particle smoothing, including ancestor-sampling methods, supplies established trajectory-sampling machinery [4]. The WPM record can attach pocket landscapes and provenance to each candidate without collapsing its identity.

A conceptual procedure is: infer a boundary distribution from the entire later sequence; propose predecessor paths under the transition model; score complete paths against all observations and anchors; preserve distinct labels through resampling; and return weighted paths or sets with a defined uncertainty interpretation. Proposal probabilities must be included in importance corrections if the algorithm is intended to sample a posterior. Heuristic scores must be labeled as such.

Do not average incompatible paths and call the mean a reconstructed physical history. For symmetric roots $\pm r$, the mean is zero, which satisfies neither branch. Report separately the

selected-path estimate, whether any retained path is close under a chosen metric, and a simultaneous path set. A broad set can retain a candidate without identifying it. If the earlier dynamical law may differ from the later law, include that uncertainty rather than claiming the later fit determines it.

12 Finite observations do not identify unrestricted equations

Theorem 12.1 (Finite-sample function ambiguity)**.** *Let distinct times $t_1, \dots, t_n$ and an interpolating function f be given. Without a restriction on the admissible function class, infinitely many functions match the same observed values and all their deterministic pocket summaries.*

Proof. For any real a, set $g_a(t) = f(t) + a\prod_{j=1}^{n}(t - t_j)$. Every added term vanishes at the observed times. Different a give different polynomials outside those points, while the observed data vector and every deterministic summary of it remain unchanged. □

The argument applies even to noiseless observations and smooth alternatives. Restricting to a degree-bounded polynomial, an identifiable differential equation or a finite grammar changes the problem and can permit recovery. Such a restriction must be declared before calling a returned function the source equation. Good fit inside a finite window does not select a unique continuation outside it.

An unknown observation map compounds this freedom with the source reparameterizations proved earlier. For that reason a complete recovery record should name both the latent source class and the observation class. "No source equation supplied" can still mean known self-dynamics, bounded coefficients, a supplied operator bank and calibrated sensors. These assumptions are mathematically substantive.

13 A linear observability trail for missing-past recovery

Consider noiseless linear dynamics $x_{j+1} = Ax_j$ and observations $y_j = Cx_j$. Let the unknown initial state be x_{-H}. Observations from time 0 to T satisfy

$$\begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_T \end{bmatrix} = \underbrace{\begin{bmatrix} CA^H \\ CA^{H+1} \\ \vdots \\ CA^{H+T} \end{bmatrix}}_{\mathcal{O}_{H,T}} x_{-H}. \tag{13.1}$$

The missing state is uniquely recoverable exactly when $\mathcal{O}_{H,T}$ has full column rank. This follows from the same nullspace argument used for source coefficients. With observation noise, small singular values determine sensitivity even when formal rank is full.

The usual boundary-state observability matrix $[C; CA; \dots; CA^T]$ can have full rank while $\mathcal{O}_{H,T}$ does not, because A^H may erase earlier directions. Thus observing the entire later boundary state accurately need not reconstruct the earlier state. If A is invertible and the boundary-state matrix has full column rank, their product with A^H retains full column rank; this is one sufficient recovery setting.

For a memory-augmented state, additional measured channels may retain the erased directions. Mathematically the state must include those memory variables. Two histories that merge only in

a measured scalar have not necessarily merged in the complete augmented state. This explains why a delayed signed trace can resolve an apparent ambiguity without violating causality.

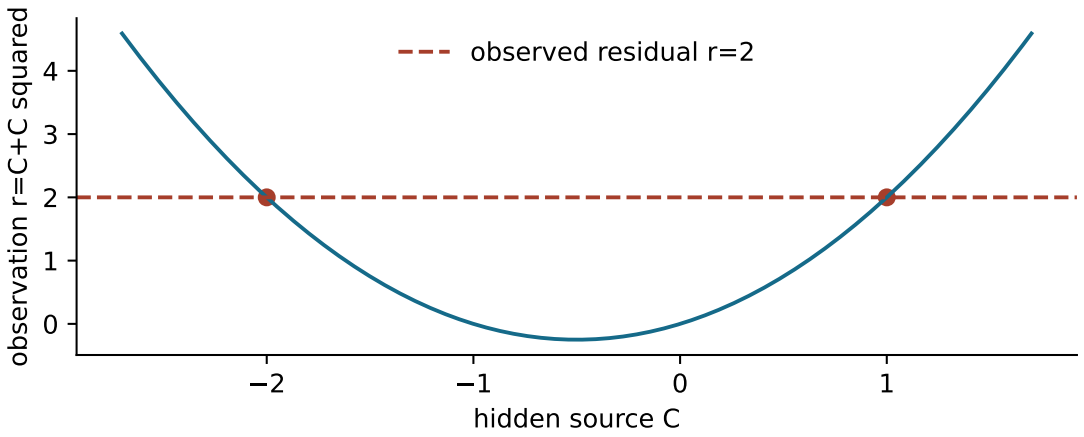


Figure 1: The calibrated observation map $r = C + C^2$ has two source values at $r = 2$. More accurate measurement of r does not choose between the two exact roots. A domain restriction or an independent informative channel is required.

14 Uncertainty sets, Bayesian branches and non-Bayesian scores

Three common outputs should not share one label. A deterministic feasible set consists of all candidates satisfying declared equations and error bounds. A Bayesian posterior is a probability law conditional on a likelihood and prior. A score-ranked list orders candidates under a chosen objective. None can be substituted for another without further assumptions.

For bounded observation error $\|Y - \mathcal{O}(x)\| \leq \delta$, define $\mathcal{C}_\delta = \{x : \|Y - \mathcal{O}(x)\| \leq \delta\}$. This is a valid inverse feasible set under that bound. Its size depends on observation conditioning and constraints. It has no nominal 90% coverage unless the error bound has that probabilistic meaning. For a posterior, branch odds follow the likelihood and prior. For score weights $w_k \propto e^{-S_k/2}$, probability interpretation requires S to arise from the appropriate log-density including priors, normalization and candidate-generation effects.

A pointwise interval at each historical time is also not automatically a simultaneous history tube. A path-set event is $\{X_{-H:-1} \in \mathcal{C}\}$; coordinatewise events form an intersection. If each coordinate interval has marginal coverage $1 - \alpha$, the full-path coverage need not equal $1 - \alpha$. Bonferroni intervals at $1 - \alpha/H$ give a union-bound construction under valid coordinatewise statements, while joint modeling can be sharper.

For a finite branch posterior, an explicit reporting rule can retain the smallest set of branches whose total posterior mass is at least $1 - \alpha$. The mass statement is conditional on that complete model, not a frequentist certificate of correctness under unmodeled dynamics. Include an inadequacy or abstention status when the candidate grammar cannot explain the observed evidence.

15 Choosing the next informative measurement

Once an ambiguity is diagnosed, use it to choose what to observe. If two candidates predict identical values on one channel but different values on another permitted channel, the second

channel can be informative. If all future observable predictions are identical, collecting more of those same observables cannot distinguish them.

In a local Gaussian coefficient model with information $\mathcal{I} \succ 0$, one proposed independent scalar measurement with sensitivity vector v and noise variance σ^2 changes information to $\mathcal{I} + vv^\top/\sigma^2$. The determinant lemma gives

$$\det(\mathcal{I} + vv^\top/\sigma^2) = \det(\mathcal{I}) \left(1 + \frac{v^\top \mathcal{I}^{-1} v}{\sigma^2}\right). \tag{15.1}$$

The formula follows by factoring $\mathcal{I}$ and applying $\det(I_p + ab^\top) = 1 + b^\top a$. It motivates measuring uncertain directions, subject to cost and the validity of the local Gaussian approximation. If the current information is singular, first inspect its nullspace rather than using an inverse that does not exist.

This design calculation distinguishes independent repeated measurements from new excitation. Both can help in different circumstances, but neither licenses an unrestricted equation claim. The final model must still be expressed in the original target units, with the observation assumptions and unresolved branches visible.

16 A recovery decision guide

Query	Construction	Required check
Known additive baseline	Fit Y minus supplied sources	Baseline error and omitted interactions
Calibrated third-source gain	Invert gain or fit raw observation map	Nonzero gain and inverse conditioning
Polynomial/nonlinear mixing	Keep all admissible branches	Global aliases, domain and channel Jacobian
Signed pocket inversion	Fit projected dictionary plus anchors	Joint rank and transformed covariance
Known reference family	Recover shared latent coordinate	Calibration, branch and sampling invertibility
Unknown delay/operator	Compare explicit candidate designs	Excitation and observational equivalence
Changing relation	Temporal basis or state-space coefficients	Local rank, drift and forecast of coefficients
Missing past	Full-path smoothing with memory channels	Model transfer and retained branch uncertainty

Every route should return the explicit equation, assumptions, units, domain, coefficient uncertainty where justified, and an ambiguity statement. If unrestricted recovery is impossible, return the identifiable combination rather than a forced single source.

17 Exercises and proof trails

Exercise 17.1. If Q annihilates constants and $C(u) = a + bu$, can the packet identify a? What extra observation is sufficient once b is known?

Solution. No: $Q(C + c) = QC$. One raw value $C(u_0)$ gives $a = C(u_0) - bu_0$. If b is also annihilated and unknown, a second independent anchor is required.

Exercise 17.2. For $Y = A+B+\lambda C$, exhibit two different latent scales with identical observations when λ is unknown.

Solution. Take $C' = 2C$ and $\lambda' = \lambda/2$. Their product is identical at every time, so pockets and the complete Y sequence cannot distinguish them without scale calibration.

Exercise 17.3. An equation has delay τ and unknown gain g acting on $B(t) = e^{2t}$. Show the identifiable quantity.

Solution. $gB(t - \tau) = ge^{-2\tau}e^{2t}$. Only $ge^{-2\tau}$ is identified by this signal. A separately known gain or a richer input is necessary for delay identification.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. The archival DOI for this study is 10.5281/zenodo.23783431.

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